SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning,Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure B
Analytical Problem Solving

Criteria	Problem Understanding *(2)*	Method Selection *(2)*	Solution Process *(2.5)*	Accuracy of Calculation *(2)*	Interpretation of Results ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I _M A Surya Kiran(192411236)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:M A SURYA KIRAN

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1.) Given the sorted data

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 30, 33, 33, 35, 35

35, 35, 36, 40, 45, 46, 52, 70

27 observations

a.) Mean & Median

Mean $= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{809}{27}$, $= 29.96 \approx 30$

Median $= \frac{27+1}{2} = 14^{th}$ value median $= 25$

b.) Mode and Modality

Modes $= 25$ and 35

= Bimodel (As only 2 values are repeating)

c.) Midrange

$\frac{Max + Min}{2} = \frac{13+70}{2} = \frac{83}{2} = 41.5$

d.) Q_1 = 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25

$Q_1 = 20$

Q_3 = 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46

$Q_3 = 35$

5 number summery

13, 20, 25, 35, 70

Min Q_1 Median Q_3 M

i) Box plot

18, 22, 25, 42, 28, 43, 33, 35, 56, 28, standardize the variable by the following:

Compute the mean absolute deviation of age.

$$\begin{array}{r} 170 \\ 76 \\ \hline 246 \end{array}$$

$$\text{Mean} = \frac{330}{10} = 33$$

15, 11, 8, 9, 5, 10, 0, 2, 23, 5

Mean deviation:

$$= \frac{15+11+8+9+5+10+2+23+5}{10}$$

$$= \frac{88}{10} = 8.8$$

(i) |18|15 | |22|11 | |25|8 | |42|9

~~= 15+~~

Mean = 33

MAD = 8.8

Suppose that the data for analysis includes the attribute age. The age values for the data tuples are.

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

Step: 1 Divide the data into bins of depth 3

Bin	original vals.	Mean smoothed values
B_1	13, 15, 16	$\frac{13+15+16}{3} = 14.67$
B_2	16, 19, 20	18.33
B_3	20, 21, 22	21
B_4	22, 25, 25	24
B_5	25, 25, 30	26.67
B_6	33, 33, 35	33.67
B_7	35, 35, 35	35
B_8	36, 40, 45	40.33
B_9	46, 52, 70	56

Replace the mean for all three value in the Bin

200, 300, 400, 600, 1000

a.) Min-Max Normalization

$$V' = \frac{V - min(A)}{max(A) - min(A)}$$

Here,

min = 200

Max = 1000

Range = 1000 - 200

= 800

original value	Normalized value
200	$\frac{200-200}{800} = 0$
300	$\frac{(300-200)}{800} = 0.125$
400	$\frac{200}{800} = 0.250$
600	$\frac{400}{800} = 0.500$
1000	$\frac{800}{800} = 1.00$

min max Normalized Data

0.00, 0.125, 0.250, 0.500, 1.000

(b) Z - Score Normalization

$$Z = \frac{x - \mu}{\sigma}$$

$\mu = \cancel{20}\ \frac{2500}{5} = 500$

standard Dev (σ)

$$\sigma = \sqrt{\frac{(200-500)^2 + (300-500)^2 + (400-500)^2 + (600-500)^2 + (1000-500)^2}{5}}$$

$= \sqrt{80000} \approx 282.84$

original val	z-score
200	-1.06
300	-0.71
400	-0.35
600	0.35
1000	1.77

z-score ND :-

-1.06, -0.71, -0.35, 0.35, 1.77

a.) min

0.000, 0.125, 0.250, 0.500, 1.000

* Given Data

5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215

a.) Equal - frequency partitioning

12 values and 3 bins

so each bin contains.

Bin	Values.
B_1	5, 10, 11, 13
B_2	15, 35, 50, 55
B_3	72, 92, 204, 215

b) Equal - width partitioning

formula:

$$\text{Bin width} = \frac{\text{Max} - \text{Min}}{\text{No. of Bins}}$$

- Min = 5
- Max = 215

$$\frac{215-5}{3} = \frac{210}{3} = 70$$

Bin intervals:

B_1 : 5 - 75

B_2 : 75 - 145

B_3 : 145 - 215

Bin	values
B_1 (5-75)	5, 10, 11, 13, 15, 35, 50, 55, 72
B_2 (75-145)	92
B_3 (145-215)	204, 215

c.) Clustering partitioning

cluster	values.
C_1	5, 10, 11, 13, 15
C_2	35, 50, 55, 72, 92
C_3	204, 215

#.

age	% fat
23	9.5
23	26.5
27	7.8
27	17.8
39	31.4
41	25.9
47	27.4
49	27.2
50	31.2

age	52	54	54	56	57	58	58	60	[illegible]
% fat	34.6	42.5	28.8	33.4	30.2	34.1	32.	41.2	[illegible]

a.) Mean, median and SD

Variable	mean	Median	SD
Age	46.44	51	13.22
%. fat	28.78	30.70	9.25

Ak 25/7/24

B) Box plot

Age : Min = 23 , Max = 61 , Median = 51

%. fat : Min = 7.8 , Max = 42.5 , 9.25